02LQD Specification and Simulation of Digital Systems

2023/02/09

1. Combinational circuit design (2 points)

Design a four-input *priority arbiter*, a circuit which accepts four inputs and outputs the index of the input with the highest value. Inputs are pure binary numbers on 8 bits. In the event of a tie, the circuit outputs the lowest index that has the highest value.

Example:

- suppose the four inputs are 28, 32, 47, and 19. The arbiter will output 2 because input 2 has the highest value, 47
- suppose the four inputs are 17, 23, 19, 23, the arbiter will output 1 because, of the two inputs (1 and 3) with the high value of 23, input 1 has the lowest index.

2. FSM design (12 points)

Design a sequential parity detector. Data arrives at a single input X, with one new bit per clock cycle. The data is grouped into packets of four bits, where the fourth bit is a parity bit. The system checks even parity: if there is an odd number of 1s in the first three bits, the fourth bit must be a 1. If an incorrect parity bit is detected, an error signal ERR is asserted. Reset is synchronous.

Design a **Mealy** FSM with X as input and ERR as output.

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with **1 process** (**4 points**).
3. Design the FSM resorting in Verilog with **2 always blocks** (**4 points**).

3. FSM-D design (16 points)

Population count, also known as popcount or Hamming weight, is the number of 1s in a data word. Design an FSM-D to compute the popcount of 8 bit data words din. When a new data word is available, the dinValid input signal is asserted and popcount computation starts. No more data words are taken into account until computation is over and the dinReady output signal is negated. When computation is over, the popcount is output on the popCnt output, the popCntValid output signal is asserted and the FSM-D signals it is ready to accept a new data word by asserting the dinReady output signal. Reset is synchronous.

1. Define the appropriate size for the popCnt output (**2 points**)
2. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)
3. Design a FSM-D:
   - create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**)
   - create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

  no later than Sunday February 12 11.59 pm
- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload ⇒ no correction ⇒ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.